QP CODE: 25900117

MAHATMA GANDHI UNIVERSITY,KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATIONS, MARCH 2025
SECOND SEMESTER

Discipline Specefic Core Course (DSC)- MG2DSCIAM100 – INTELLIGENT AUTOMATION TECHNIQUES
(2024 Admission Onwards)

Duration:1.5 Hrs **Maximum Marks:50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)
Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

ANSWER KEY

PART A – Multiple Choice Questions

Answer all questions. Each question carries 1 Mark

		Learning Domain	CO No:	Marks
1	d) Reinforcement Learning	K	CO1	1
2	d) pd.read_sql()	U	CO2	1
3	d) Logistic Regression	K	CO3	1
4	c) Nodes and weights only	U	CO1	1
5	d) Simple and easy to implement	K	CO2	1
6	d) All of the above	U	CO1	1
7	d) To make the convergence and gradient larger, reduce oscillations	K	CO2	1
8	c) Learning through rewards and penalties	U	CO3	1
9	a) Supervised Learning	K	CO3	1
10	d) All of the above	An	CO1	1

PART B – Short Answer Questions

Answer 4 questions. Each question carries 5 Marks

	Learning Domain	CO No:	Marks
11.Real-world Applications of AI: **• Healthcare** **• Finance** **• Retail** **• Transportation** **• Agriculture**	**K**	**1×5 = 5, CO1**	**5**
12.Default File Extension for Python: **• .py**	**U**	**Direct = 5 CO2**	**5**
13.Applications of Unsupervised Learning: **• Customer segmentation** **• Anomaly detection** **• Clustering** **• Market analysis** **• Compression**	**A**	**1×5 = 5, CO3**	**5**
14.Variables in Python with Example: **• Concept – 2** **• Example – 3** **• Dynamic typing, naming rules**	**U**	**2+3 = 5, CO2**	**5**
15.Difference between Agglomerative & Divisive Clustering: **• 5 distinct differences**	**K**	**1×5 = 5, CO3**	**5**
16.ML in Financial Sector: **• Credit scoring** **• Fraud detection** **• Trading** **• Risk management** **• Churn prediction**	**A**	**1×5 = 5 CO1**	**5**

PART C – Essay Questions

Answer any 2 questions. Each carries 10 Marks

	Learning domain	CO No:	Marks
17.DB Connection in Python: **• Steps – 7** **• Code – 3**	**K**	**7+3 = 10 CO2**	**10**

e.g., sqlite3 connect, execute, fetch

18.Regression Analysis: **• Theory – 6** **• Code – 4** **e.g., sklearn LinearRegression**	**K**	**6+4 = 10**	**CO3**	**10**
19.Compare Learning Types: **• Supervised, Unsupervised, Reinforcement** **• Theory – 6** **• Examples – 4**	**An**	**6+4 = 10**	**CO1**	**10**
20.Debugging in Jupyter Notebook: **• 7 methods (print, %debug, pdb, etc.)** **• Example – 3**	**U**	**7+3 = 10**	**CO2**	**10**